# Practical - 5

## Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance using Regular Expressions.

## Theory :

### 1. Are correct input types and attributes used?

Yes. The form uses suitable HTML5 input types: type="text" for name, type="email" for email, type="tel" for mobile number, type="password" for both password fields, select for course and year, radio buttons for gender, and a checkbox for terms. Placeholders guide the user, and each field has a unique id for JavaScript access.

### 2. Are validation errors displayed near the relevant fields?

Yes. A small element is placed directly below each field (for example nameError and emailError). When validation fails, JavaScript writes a clear message such as "Please enter a valid email." in red inside that element. All old messages are cleared on every submit, so only current errors are shown.

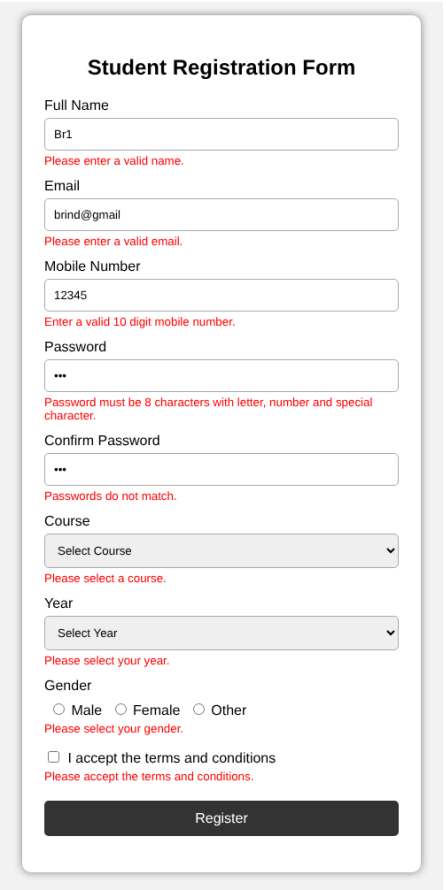
### 3. Is password strength checked?

Yes. A regular expression, /^(?=.*[A-Za-z])(?=.*[0-9])(?=.*[@#$%]).{8,}$/, requires at least 8 characters with a letter, a number, and a special character (@ # $ %). The confirm password field is also compared with the password to make sure both match.

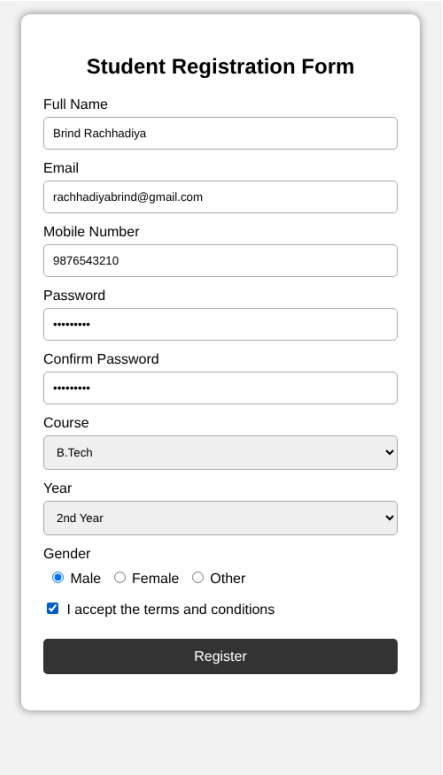
### 4. Is the form accessible using keyboard and screen-reader-friendly labels?

Yes. Standard form elements are used, so the user can move between fields with the Tab key, choose options with arrow keys, tick the checkbox with Space, and submit with Enter. Every field has a visible label above it and an error message below it. Linking each label to its input with the for attribute improves screen-reader support further.

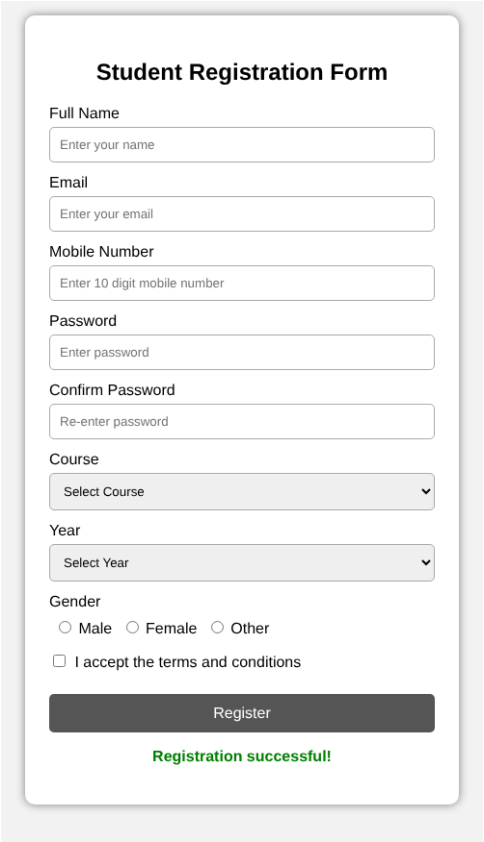
## Implementation/ Output/Result


Student Registration Form
Full Name
Br1
Please enter a valid name.
Email
brind@gmail
Please enter a valid email.
Mobile Number
12345
Enter a valid 10 digit mobile number.
Password
Password must be 8 characters with letter, number and special character.
Confirm Password
Passwords do not match.
Course
Select Course
Please select a course.
Year
Select Year
Please select your year.
Gender
Male Female Other
Please select your gender.
I accept the terms and conditions
Please accept the terms and conditions.
Register

1


Student Registration Form
Full Name
Brind Rachhadiya
Email
rachhadiyabrind@gmail.com
Mobile Number
9876543210
Password
Confirm Password
Course
B.Tech
Year
2nd Year
Gender
Male Female Other
I accept the terms and conditions
Register

2




3

Figure: (1) invalid input showing error messages, (2) valid input filled in, (3) successful registration message.

## GitHub Repository Link:

https://github.com/rachhadiyabrind-del/WDF_B/tree/main/Practical%205

## Learning Outcomes:

- Build forms using HTML5 input types and attributes
- Validate user input with JavaScript and regular expressions
- Display clear error messages near the relevant fields
- Check password strength and confirm password matching
- Design forms that are usable with keyboard and labels
- Test the form with valid and invalid sample inputs

## Faculty Remarks: (Keep this blank for faculty)